

Gender Classification by Facial Attribute Using SVM

An end-to-end facial attribute classification system combining transfer-learned ResNet-18 deep feature extraction with a maximum-margin Support Vector Machine classifier on the UTKFace benchmark.

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Project Overview & Core Motivation

Decoupled deep representation learning and convex maximum-margin classification.



Project Objective Build an automated, robust facial gender classification pipeline that ingests raw facial imagery across broad demographic spectrums, standardizes spatial context, extracts compact discriminative embeddings, and delivers high generalization accuracy.

THE TECHNICAL CHALLENGE

- **Morphological Variance**
Extreme non-linear diversity driven by age, ethnicity, illumination, and head orientation.
- **Subtle Dimorphism**
Secondary sexual characteristics (jawline contour, brow prominence, cheekbone geometry) require sensitive spatial representations.
- **Classical Limitations**
Handcrafted descriptors (HOG, LBP, Haar cascades) degrade under unconstrained pose, lighting, and expression variations.

THE PROPOSED SOLUTION

- **Deep Feature Extraction**
Transfer-learned ResNet-18 transforms 224×224 RGB faces into compact 512-dimensional bottleneck embeddings.
- **Maximum-Margin Boundary**
RBF-kernel SVM finds a globally optimal separating hyperplane, avoiding local minima inherent to pure cross-entropy backprop.
- **Decoupled Modularity**
Allows independent feature-space normalization, disciplined regularization, and low-latency (<35ms) inference.

Support Vector Machine: Mathematical Foundation

Algorithmic formulation, structural risk minimization, and decision boundary mechanics.

Definition A Support Vector Machine (SVM) is a supervised discriminative statistical learning algorithm that constructs an optimal separating hyperplane in high-dimensional feature space by maximizing the geometric margin the minimum perpendicular distance between the hyperplane and the closest training samples of each class.

1

Structural Risk Minimization

Enforces margin maximization to minimize an upper bound on true generalization error rather than solely minimizing empirical training error.

2

Convex Global Optimum

Cast as a convex Quadratic Programming (QP) problem, guaranteeing that any local solution is the unique global optimum.

3

Support Vector Sparsity

The decision boundary is defined strictly by critical points lying on the margin; interior non-boundary samples can be removed without altering the hyperplane.

SVM Geometry: Hyperplane & Margin Derivation

Primal quadratic optimization formulation for linearly separable embeddings.

$$\min(w, b) \quad \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1, \quad \forall i \in \{1, \dots, N\}$$

- **Hyperplane Equation**

$w^T x + b = 0$ (normal weight vector w , scalar bias b).

- **Canonical Margin Planes**

$w^T x + b = +1$ (Male boundary) and $w^T x + b = -1$ (Female boundary).

- **Geometric Margin**

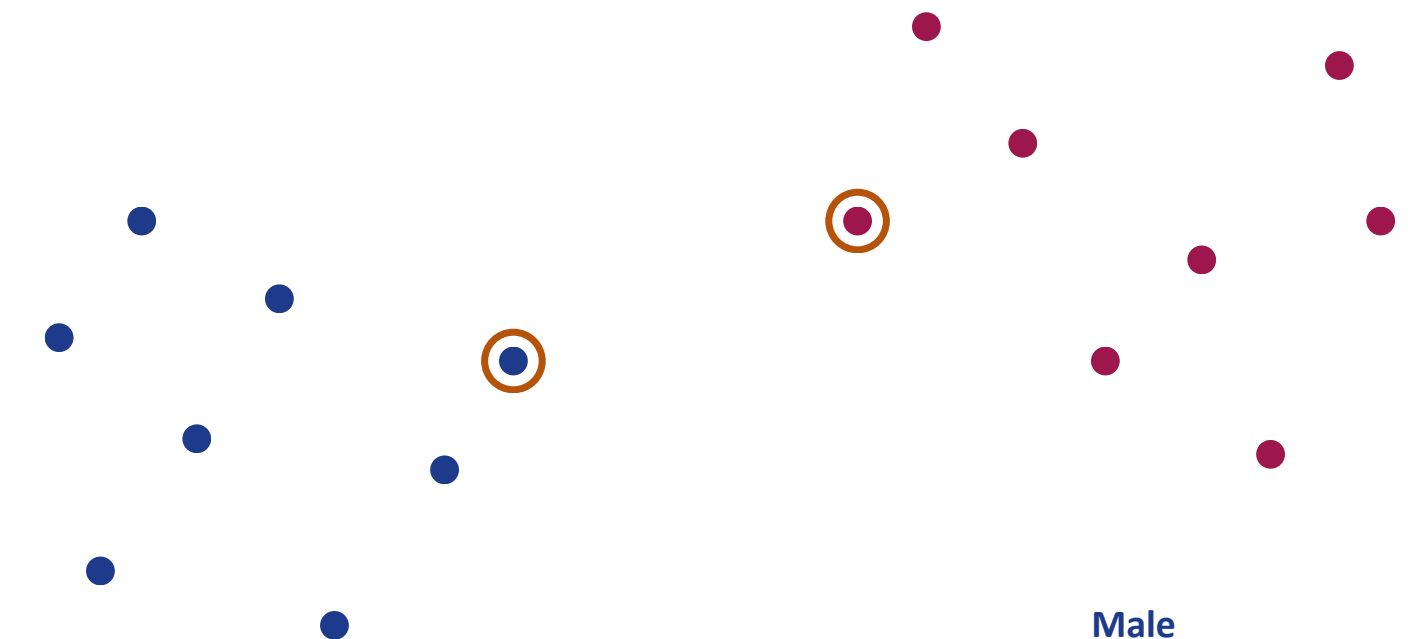
The physical width separating canonical planes is $\gamma = 2 / \|w\|$. Maximizing width is equivalent to minimizing $\frac{1}{2} \|w\|^2$.

- **Dimension Robustness**

Generalization capacity is governed by margin width rather than input feature dimensionality ($d = 512$).

CONCEPTUAL MARGIN PLOT

Female



○ support vectors

Soft-Margin Optimization & Slack Variables

Handling demographic overlap, label noise, and edge cases via Cortes–Vapnik formulation.

$$\min(w, b, \xi) \quad \frac{1}{2} \|w\|^2 + C \sum_{i=1 \dots n} \xi_i \quad \text{s.t.} \quad y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

1

Role of Slack (ξ_i)

Quantifies margin violations. $\xi_i = 0$ (correct & outside margin), $0 < \xi_i \leq 1$ (correctly classified inside margin band), $\xi_i > 1$ (misclassified).

2

Regularization Parameter (C)

Controls the trade-off between maximizing margin width and penalizing slack violations. Large C enforces hard constraints (overfitting risk); small C tolerates violations to preserve a wider, more generalizable boundary.

3

Empirical Finding (C = 0.01)

Grid search demonstrated C = 0.01 achieved peak validation performance, preventing overfitting on ambiguous infant and elderly facial features.

Non-Linear Boundaries & The Kernel Trick

Dual formulation and non-linear feature projection using the Radial Basis Function (RBF) kernel.

$$\max(\alpha) \quad \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \quad \text{where} \quad K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$



Implicit High-Dimensional Mapping

The kernel trick computes inner products in infinite-dimensional Hilbert space without explicitly computing non-linear coordinates $\phi(x)$.



RBF Decision Contours

Evaluates Gaussian Euclidean-distance similarity, establishing smooth localized decision envelopes around support vectors.



Scale Bandwidth ($\gamma = \text{'scale'}$)

Automatically adapts bandwidth as $\gamma = 1 / (d \cdot \text{Var}(X))$, standardizing kernel behavior across all 512 dimensions.

Dataset: UTKFace Benchmark & Partitioning

Demographic breakdown, label encoding, and stratified data splits.

24102

Total Face Images

52.2% / 47.8%

Male (12,581) vs. Female (11,521)

[age]_[gender]_[race]_[dt].jpg

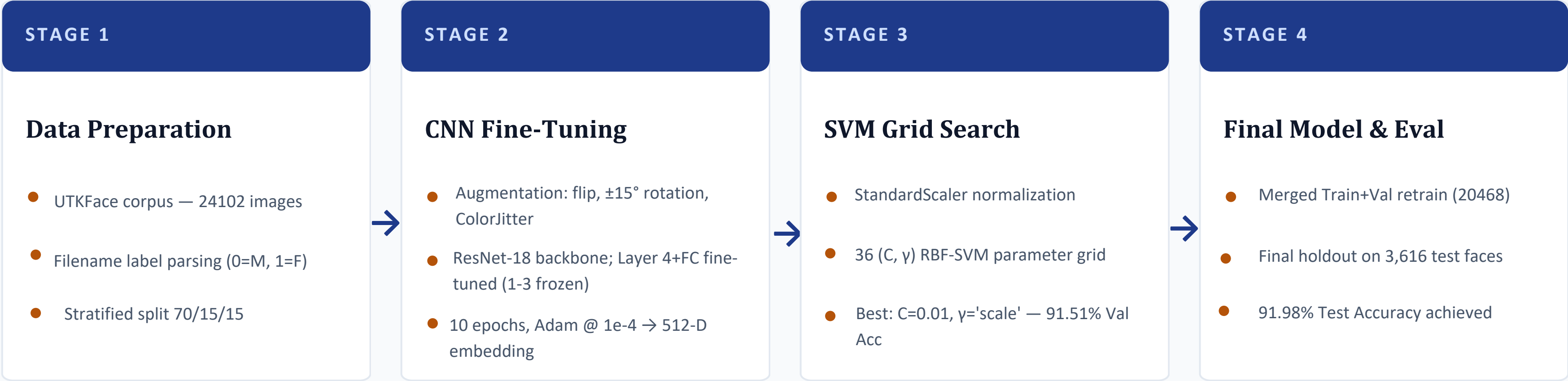
Gender token: 0 = Male, 1 = Female

STRATIFIED PARTITIONING

Partition	Share	Count	Purpose
Training Set	70.0%	16,871 faces	Fine-tuning ResNet-18 Layer 4 + FC; preliminary SVM fit
Validation Set	15.0%	3,615 faces	Grid search hyperparameter optimization across 36 (C,γ) pairs
Test Set (Holdout)	15.0%	3,616 faces	Untouched benchmark evaluation
Merged Retraining Set	85.0%	20,486 faces	Train + Val merged fit before final holdout testing

System Training Pipeline Architecture

Sequential four-stage pipeline: data preparation, CNN fine-tuning, SVM grid search, and final retraining.



Feature Extractor: ResNet-18 Backbone & GAP

Layer freezing rationale, residual skip connections, and Global Average Pooling vector generation.

Core Rationale Residual skip connections $y = F(x) + x$ resolve vanishing gradients. Freezing early layers preserves general low/mid-level visual filters while unfreezing Layer 4 tailors high-level representations to facial dimorphism.

Stage	Output	Represents
Input	Normalized 3×224×224 RGB crop (Resize 256 → CenterCrop 224)	—
Conv1 + MaxPool (Frozen)	64 ch, 56×56 map	Edges, gradients, low-level textures
Layers 1–3 (Frozen)	256 ch, 14×14 map	Mid-level contours, eye sockets, nose bridge
Layer 4 (Fine-Tuned)	512 ch, 7×7 map	High-level dimorphism: jaw, brow, cheekbones
AdaptiveAvgPool2d (GAP)	1×512 vector	Spatially invariant embedding → Scaler & SVM

Shaded rows = fine-tuned / trainable components

RBF-SVM Grid Search & Hyperparameter Tuning

Exhaustive validation across 36 hyperparameter pairs and retraining protocol.

36

Grid Combinations (6×C × 6×γ)

C = 0.01

Optimal Regularization

'scale'

Optimal Kernel Bandwidth (γ)

91.51%

Best Validation Accuracy



Search Space

C ∈ {0.01, 0.1, 1, 10, 100, 1000} and γ ∈ {'scale','auto',0.001,0.01,0.1,1}, evaluated strictly on 3,556 validation faces to prevent data leakage.



Why Small C Won

Larger values of C forced the model to memorize noisy demographic boundaries (infants and elderly), reducing generalization. C=0.01 established smooth, highly regularized margins.

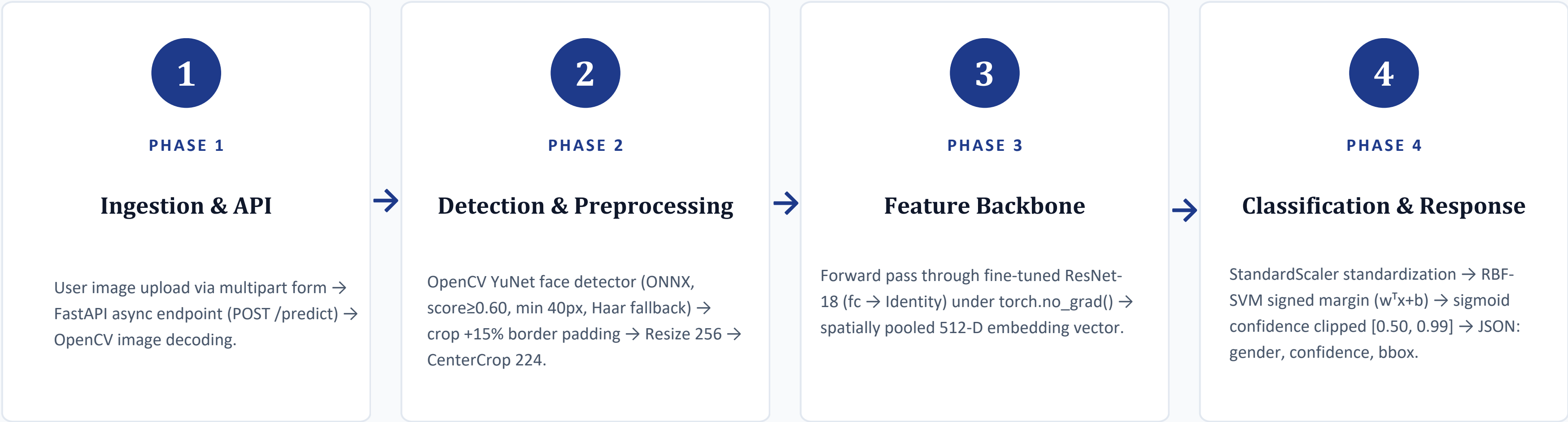


Retraining Protocol

Following parameter discovery, the SVM was refit on the unified 85% corpus (20,151 samples) to maximize support vector density prior to holdout testing.

Production Real-Time Inference Pipeline

Four operational stages from image ingest to calibrated JSON prediction.



Experimental Results & Evaluation Metrics

Quantitative classification benchmarks on the untouched 15% holdout test partition (3,616 faces).

91.98%

Test Accuracy

3,225 / 3,616 faces correctly classified

Only 285 total errors across the holdout test partition

PERFORMANCE BY CLASS

Class	Precision	Recall	F1-Score	Support
Male (0)	91.5%	93.0%	92.2%	1,859
Female (1)	92.5%	90.9%	91.7%	1,698
Overall Model	—	—	—	Acc: 91.98%

- ✓ Balanced high-precision classification across both genders with less than 2% recall difference.
- ✓ Strong margin separation confirmed by balanced F1-scores (92.2% Male vs. 91.7% Female).

PROJECT DEFENSE & CONCLUSION

Questions & Technical Discussion

Thank you for your time. We welcome questions regarding our project.

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